

Math 196 Lecture Notes: Week 7

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0.1 Orthonormal Bases

Recall that a list of vectors $u_1, \dots, u_r$ is said to be *orthonormal* if

$$u_i \cdot u_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

Such vectors are always linearly independent. Indeed, given a linear combination

$$\sum_i x_i u_i = v$$

we have

$$x_j = u_j \cdot v$$

and so if $v = 0$, all $x_j = 0$. If $u_1, \dots, u_r$ span W a subspace, they are called an orthonormal basis for W .

0.2 Orthogonal Projection

Given an orthonormal basis for W , any vector $v \in \mathbb{R}^n$ can be decomposed as $v_1 + v_2$, where

$$v_1 = \sum_i (v \cdot u_i) u_i \in W$$

$$v_2 = v - v_1 \in W^\perp$$

Such a decomposition as an element of W and an element of $W^\perp$ is unique, and we call v_1 the projection of v onto W .

0.3 Graham-Schmidt Process

To compute this in general, we had a recursive process that replaces the vectors in a basis $v_1, \dots, v_r$ for W with new vectors $u_1, \dots, u_r$, one at a time, so that $u_1, \dots, u_r$ is an orthonormal basis for W . Informally, at each step, we use the previously constructed $u_1, \dots, u_i$ to make v_{i+1} orthogonal to all of them, then normalize it. Once we have the orthonormal basis, we can take

$$U = \begin{bmatrix} \vdots & & \vdots \\ u_1 & \dots & u_r \\ \vdots & & \vdots \end{bmatrix}$$

and then projection is given by the matrix $UU^\top$.

1 Algebraic Properties of Transpose

The transpose operation has some algebraic properties that will be useful to us.

1. $(A + B)^\top = A^\top + B^\top$
2. $(cA)^\top = cA^\top$
3. $(A^\top)^\top = A$
4. $(AB)^\top = B^\top A^\top$

The first three of these are straightforward to check from the definition. One can think of (1) and (2) as saying that transpose is linear as an operation on matrices. The fourth one is a bit more interesting. One way to see it is to divide A into rows and B into columns:

$$A = \begin{bmatrix} \dots & r_1 & \dots \\ & \vdots & \\ \dots & r_m & \dots \end{bmatrix}$$

$$B = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \dots & v_n \\ \vdots & & \vdots \end{bmatrix}$$

Then the i, j entry of AB is $r_i v_j$, which is the j, i entry of $(AB)^\top$. On the other hand

$$A^\top = \begin{bmatrix} \vdots & & \vdots \\ r_1^\top & \dots & r_m^\top \\ \vdots & & \vdots \end{bmatrix}$$

$$B^\top = \begin{bmatrix} \dots & v_1^\top & \dots \\ & \vdots & \\ \dots & v_n^\top & \dots \end{bmatrix}$$

and so $B^\top A^\top$ has j, i entry $v_j^\top r_i^\top = r_i v_j$.

In particular, for the projection matrix $P = UU^\top$, we have $P^\top = UU^\top = P$. This is to say that P is a *symmetric matrix*, its entries a_{ij} satisfy $a_{ij} = a_{ji}$.

2 Orthogonal Matrices

Suppose that $u_1, \dots, u_n$ is an orthonormal basis for $\mathbb{R}^n$. We can package these into a matrix

$$A = \begin{bmatrix} \vdots & & \vdots \\ u_1 & \dots & u_n \\ \vdots & & \vdots \end{bmatrix}$$

Because we started with a basis, this matrix is invertible. We can say more in this case. We will first show that A preserves the dot product, in the sense that

$$(Av) \cdot (Aw) = v \cdot w$$

for any vectors $v, w \in \mathbb{R}^n$. To see this, let

$$v = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad w = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Then

$$v \cdot w = \sum_i x_i y_i$$

On the other hand, we have

$$Av = x_1 u_1 + \cdots + x_n u_n$$

$$Aw = y_1 u_1 + \cdots + y_n u_n$$

and so, by the distributive property for dot product

$$(Av) \cdot (Aw) = \sum_{i,j} x_i y_j (u_i \cdot u_j)$$

now $u_i \cdot u_j$ is 0 if $i \neq j$, and 1 if $i = j$, so this comes out to

$$\sum_i x_i y_i = v \cdot w$$

As a consequence of A preserving the dot product, A preserves length, in the sense that $|Av| = |v|$ for any v . Moreover, A preserves orthogonality, in the sense that if v and w are orthogonal, so are Av and Aw .

In fact, any matrix A which preserves length also preserves orthogonality, because it must take the right triangle with sides $v, w, v + w$ to a right triangle with sides $Av, Aw, Av + Aw$. Such a matrix must take the standard basis $e_1, \dots, e_n$ to another orthonormal basis $u_1, \dots, u_n$. These u_i are the columns of A so we have come full circle.

Conclusion. *The following are equivalent, for A an $n \times n$ matrix:*

1. *The columns of A are an orthonormal basis.*
2. *$|Av| = |v|$ for all v .*
3. *$Av \cdot Aw = v \cdot w$ for all v, w .*

Such a matrix is said to be *orthogonal*. In $\mathbb{R}^2$, we have seen that rotation matrices

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

are orthogonal. Thinking geometrically, reflection across any line through the origin preserves the length of vectors, so these are orthogonal as well. It turns out that the 2×2 orthogonal matrices are exactly the rotations and reflections. In $\mathbb{R}^3$, rotation about any axis is length preserving, so its matrix is orthogonal. The same goes for reflection across any plane through the origin. Again, these are all the orthogonal 3×3 matrices.

Another useful thing about orthogonal matrices is that they turn out to be easy to invert:

Proposition. *If A is orthogonal, then $A^\top = A^{-1}$.*

Proof. For any matrix A with columns $v_1, \dots, v_n$, the i, j entry in $A^\top A$ is $v_i^\top v_j = v_i \cdot v_j$. If A is orthogonal, this is 0 for $i \neq j$ and 1 for $i = j$, so $A^\top A = I_n$. Multiplying on the right by A^{-1} , we get $A^\top = A^{-1}$. $\square$

In fact, the condition that $A^\top = A^{-1}$ is equivalent to the other properties characterizing orthogonal matrices.

Proposition. *If A and B are orthogonal, then AB and A^{-1} are orthogonal.*

Proof. We check that AB and A^{-1} are length preserving:

$$|ABv| = |A(Bv)| = |Bv| = |v|$$

first using the fact that A is length preserving, then that B is length preserving. Similarly,

$$|v| = |AA^{-1}v| = |A^{-1}v|$$

using the fact that A is length preserving. □

3 Least Squares

3.1 Distance and Least Squared Solutions

Given two vectors v and w , the distance between them (between their endpoints) is the length of their difference, $d(v, w) = |v - w|$. With this view of distance, we can characterize orthogonal projection in a different way.

Proposition. *Let W be a subspace of $\mathbb{R}^n$ and $v \in \mathbb{R}^n$ be a vector. Then $\text{proj}_W(v)$ is the vector $w \in W$ which is closest to v , in the sense that it minimizes $d(v, w)$.*

Proof. Let $w = \text{proj}_W(v)$ and $w' \in W$ be another vector. We want to show that

$$d(v, w') \geq d(v, w)$$

Because $v - w \in W^\perp$ and $w' - w \in W$, these vectors are orthogonal, and so we have a right triangle with corners w', w, v . [Draw picture]. The Pythagorean theorem (or alternatively an explicit computation with dot products) then tells us that

$$|w' - v|^2 = |w - v|^2 + |w' - w|^2 \geq |w - v|^2$$

and so

$$|w' - v| \geq |w - v|$$

as required. □

If we present W as $\text{im}(A)$ for some matrix A , then we can view this in the following way. An equation $Ax = v$ may be inconsistent if $v \notin \text{im}(A)$. In this case, we can try to find something which is as close to a solution as possible. One interpretation of this is to find an x so that $|Ax - v|$ is as small as possible. From the above discussion, this will be true provided that $Ax = \text{proj}_W(v)$. As $\text{proj}_W(v) \in \text{im}(A)$, this equation is always solvable.

It is equivalent to say that we are looking for an x such that $|Ax - v|^2$ is as small as possible. This is the sum of squares of the entries of $Ax - v$, so such an x is called a *least squares* solution to the linear system $Ax = v$.

From what we have seen so far, we can find a least squares solution x via the following steps:

1. Find a basis for $\text{im}(A)$, but bringing A to reduced row echelon form and picking out the pivot columns.
2. Apply the Gram-Schmidt process to this basis to find an orthonormal basis.
3. Use this orthonormal basis to compute the projection w of v onto $\text{im}(A)$.
4. Solve the system $Ax = w$.

This would be a quite laborious process, so it is quite fortunate that there is a more direct way to find least-squares solutions.

3.2 The Normal Equation

Let $Ax = v$ be a linear system. We call the equation

$$A^\top Ax = A^\top v$$

the *normal equation* associated with the original linear system.

Proposition. *A vector x is a least squares solution to $Ax = v$ if and only if it is an exact solution to the normal equation.*

Proof. From our original characterization of orthogonal projection, we know that x is a least squares solution if and only if $Ax - v \in \text{im}(A)^\perp$, as this means that $Ax = \text{proj}_{\text{im}(A)}(v)$. On the other hand, $\text{im}(A)^\perp = \ker(A^\top)$, so this holds if and only if

$$0 = A^\top(Ax - v) = A^\top Ax - A^\top v$$

□

Example 1. Consider the linear system

$$\begin{bmatrix} 1 & 0 \\ 1 & -2 \\ 1 & -1 \end{bmatrix} x = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

We apply row operations to the augmented matrix as follows:

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 1 & -2 & 1 \\ 1 & -1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & -1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & -1/2 \\ 0 & -1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & -1/2 \\ 0 & 0 & -1/2 \end{array} \right]$$

From this, we find that the system is inconsistent. We now look for a least squares solution, so we form the normal equation by applying the transpose of the matrix to both sides:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & -2 \\ 1 & -1 \end{bmatrix} x = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Computing the matrix products, this is

$$\begin{bmatrix} 3 & -3 \\ -3 & 5 \end{bmatrix} x = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Now we can solve the resulting linear system as follows:

$$\left[\begin{array}{cc|c} 3 & -3 & 1 \\ -3 & 5 & -2 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 3 & -3 & 1 \\ 0 & 2 & -1 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & -1 & 1/3 \\ 0 & 1 & -1/2 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 1/6 \\ 0 & 1 & -1/2 \end{array} \right]$$

This tells us that $x = \begin{bmatrix} -1/6 \\ -1/2 \end{bmatrix}$ is a least squares solution to the original system. The vector

$$\begin{bmatrix} 1 & 0 \\ 1 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1/6 \\ -1/2 \end{bmatrix} = \begin{bmatrix} -1/6 \\ 5/6 \\ 1/3 \end{bmatrix}$$

is the closest element of the image to $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. Here the optimal distance squared is

$$\left\| \begin{bmatrix} -1/6 \\ 5/6 \\ 1/3 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\|^2 = 1/36 + 1/36 + 4/36 = 6/36 = 1/6$$

Let's again consider the normal equation, in generality:

$$A^\top Ax = A^\top v$$

Because solving this is equivalent to solving $Ax = \text{proj}_{\text{im}(A)}v$, this system is always consistent. Moreover, if A is injective, then this system always has a unique solution. Indeed:

Proposition. $\ker(A) = \ker(A^\top A)$ for any A . In particular, if $\ker(A) = \{0\}$, then $A^\top A$ is a square matrix with kernel $\{0\}$, so is invertible.

Proof. Clearly if $Ax = 0$, then $A^\top Ax = 0$. On the other hand, if $A^\top Ax = 0$, then $Ax \in \ker(A^\top) = (\text{im}(A))^\perp$, but also $Ax \in \text{im}(A)$, so we must have $Ax = 0$. $\square$

In the case that A is injective, we can write a formula for the unique least squares solution to $Ax = v$. This is the unique exact solution to $A^\top Ax = A^\top v$, which is $x = (A^\top A)^{-1}A^\top v$. This also gives us a new formula for projection. Recalling that the least squares solution satisfies $Ax = \text{proj}_{\text{im}(A)}(v)$, we have

$$\text{proj}_{\text{im}(A)}(v) = A(A^\top A)^{-1}A^\top v$$

This makes sense provided that A is injective. Thus, one way to compute a matrix for projection onto W is to find a (not necessarily orthonormal) basis of W , then put the vectors into a matrix A . This A will be injective and have $\text{im}(A) = W$, so we have

$$\text{proj}_W = A(A^\top A)^{-1}A^\top$$

This is a generalization of the formula

$$\text{proj}_v = \frac{1}{|v|^2}vv^\top$$

for projection onto a line.

3.3 Data Fitting

We expect linear systems $Ax = v$ to be inconsistent when they have more equations than unknowns. Such systems are called *overdetermined*. One way that overdetermined systems can arise is in the problem of fitting a function with a small number of parameters to a large data set.

For example, suppose that we have some data points (x_i, y_i) , and we have good reason to believe that y is approximately a linear function of x , of the form $y = ax + b$, where a, b are unknown constants. Then each data point yields a linear equation that should be satisfied by a, b . If we have two data points, then there will be a unique line connecting them and we can find a, b by solving a linear equation.

Example 2. Suppose we have data points $(1, 2)$ and $(3, 3)$. These translate into linear equations

$$\begin{cases} a + b = 2 \\ 3a + b = 3 \end{cases}$$

Solving the system, we obtain

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 3 & 1 & 3 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & -2 & -3 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 3/2 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 1/2 \\ 0 & 1 & 3/2 \end{array} \right]$$

and so $a = 1/2$, $b = 3/2$, and we predict that $y = \frac{x}{2} + \frac{3}{2}$.

If we have more than two data points, each will contribute an equation and so the system will be overdetermined. If y is not exactly equal to $ax + b$ or is imperfectly measured, the system is likely to be inconsistent as well. In this case, a least squares solution gives us a reasonable “best guess” at the true values of a, b .

Example 3. Suppose we add to our data set the point $(-1, 0)$. Now our system is

$$\begin{cases} a + b = 2 \\ 3a + b = 3 \\ -a + b = 0 \end{cases}$$

Or equivalently,

$$\begin{bmatrix} 1 & 1 \\ 3 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$$

We know this is inconsistent: the first two equations already imply that a and b take the values obtained above, but these do not satisfy the third equation. Looking for a least squares solution, we form the normal equation

$$\begin{bmatrix} 1 & 3 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 & 3 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$$

which comes out to

$$\begin{bmatrix} 11 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 11 \\ 15 \end{bmatrix}$$

We can solve this as follows:

$$\left[\begin{array}{cc|c} 11 & 3 & 11 \\ 3 & 3 & 15 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 3/11 & 1 \\ 3 & 3 & 15 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 3/11 & 1 \\ 0 & 24/11 & 2 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 3/11 & 1 \\ 0 & 1 & 11/12 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 3/4 \\ 0 & 1 & 11/12 \end{array} \right]$$

Thus $y = \frac{3}{4}x + \frac{11}{12}$ is the line that best fits these data points, in the sense that the sum of the squares of the errors is as small as possible. [Draw picture]

This method can be applied to other families of functions as well. For example, we might expect that our y is a quadratic function of x , $y = ax^2 + bx + c$. Given a list of data points, we could then set up a system of linear equations, and a least squares solution would give us a quadratic function that minimizes the sum of squares of error over all the data points. For this to work, we just need the output of the function to depend linearly on the *parameters*, here a, b, c , so that when we plug in inputs and outputs, we get linear equations in the parameters. There is also nothing stopping us from doing this with a multi variable function, as long as everything is linear in the parameters. In general, this process is known as *linear regression*.

Example 4. Suppose you are on a distant planet with unknown gravity. You throw a stone upwards, letting it fall back to the ground. Using a camera and meter stick, you get the following data points for the vertical displacement of the stone at different times.

t	1	1.5	2	2.5
y	3.5	5	4	2.5

Our prediction for y is $y = vt - \frac{g}{2}t^2$ where v is the initial upward velocity and g is the acceleration due to gravity, both unknown. Plugging in each data point, we get linear equations that must be satisfied by v and g .

$$\begin{cases} v - 0.5g = 3.5 \\ 1.5v - 1.125g = 5 \\ 2v - 2g = 4 \\ 2.5v - 3.125g = 2.5 \end{cases}$$

Or equivalently

$$\begin{bmatrix} 1 & -0.5 \\ 1.5 & -1.125 \\ 2 & -2 \\ 2.5 & -3.125 \end{bmatrix} \begin{bmatrix} v \\ g \end{bmatrix} = \begin{bmatrix} 3.5 \\ 5 \\ 4 \\ 2.5 \end{bmatrix}$$

The normal equation is

$$\begin{bmatrix} 1 & 1.5 & 2 & 2.5 \\ -0.5 & -1.125 & -2 & -3.125 \end{bmatrix} \begin{bmatrix} 1 & -0.5 \\ 1.5 & -1.125 \\ 2 & -2 \\ 2.5 & -3.125 \end{bmatrix} \begin{bmatrix} v \\ g \end{bmatrix} = \begin{bmatrix} 1 & 1.5 & 2 & 2.5 \\ -0.5 & -1.125 & -2 & -3.125 \end{bmatrix} \begin{bmatrix} 3.5 \\ 5 \\ 4 \\ 2.5 \end{bmatrix}$$

This comes out to approximately

$$\begin{bmatrix} 13.5 & -14 \\ -14 & 15.3 \end{bmatrix} \begin{bmatrix} v \\ g \end{bmatrix} = \begin{bmatrix} 25.3 \\ -23.2 \end{bmatrix}$$

This has solution approximately

$$\begin{bmatrix} v \\ g \end{bmatrix} = \begin{bmatrix} 5.8 \\ 3.8 \end{bmatrix}$$

Thus 3.8 is a good guess for the planets gravitational acceleration.